

Mahim Katiyar

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🐙 GitHub

🌐 LinkedIn

🔗 LeetCode

Education

Thapar Institute of Engineering and Technology, Patiala

2023 – Present

B.E. in Computer Engineering — CGPA: 8.39/10

Woodbine Gardenia School, Kanpur

2022

Class 12 (CBSE) — Percentage: 95%

Woodbine Gardenia School, Kanpur

2020

Class 10 (CBSE) — Percentage: 97.8%

Skills

Programming: Python, C++, SQL

Core CS: DSA, OOP, DBMS, OS, Computer Networks

ML/DL: Feature Engineering, Model Evaluation, Hyperparameter Tuning, CNNs, RNNs, LSTMs, Transfer Learning

GenAI: RAG, LangChain, LangGraph, Agentic AI

Frameworks/Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, Streamlit, spaCy, ChromaDB, Pandas, NumPy

Backend/DB: FastAPI, Pydantic, MySQL, PostgreSQL

Tools: Docker, Git, GitHub, MS Excel

Concepts: System Design

Projects

ResearchMind – Multi-Agent AI Research System

🐙 GitHub

Python, LangChain, Groq, Tavily, BeautifulSoup, Streamlit

- Built a multi-agent research pipeline with specialized **Search, Reader, Writer, and Critic** agents to generate cited reports from arbitrary topics.
- Integrated **Tavily** and **BeautifulSoup** for web search and source extraction, grounding citations in raw tool output to reduce hallucinated sources.
- Designed the multi-agent pipeline to be topic-agnostic, allowing structured, cited reports to be generated for arbitrary research queries with no manual intervention.

AI Video Assistant

🐙 GitHub

Python, Streamlit, RAG, ChromaDB, Mistral, Deepgram, Whisper

- Built an AI video assistant that converts YouTube/audio into transcripts, summaries, action items, decisions, and open questions.
- Implemented multi-engine transcription using **Deepgram, Sarvam, and Whisper** and a **RAG chatbot** with embeddings and ChromaDB for context-aware Q&A.
- Developed persistent CLI and Streamlit interfaces with Mistral-powered summarization and per-meeting vector collections.

Python Package for TOPSIS (Multi-Criteria Decision Making)

🐙 GitHub

Python, NumPy, Pandas, CLI Packaging

- Built and packaged a reusable Python library implementing the TOPSIS algorithm following PyPI-style distribution conventions.
- Designed a CLI supporting CSV/XLSX inputs with automated validation of weights and impacts.
- Implemented weighted normalization, ideal-solution computation, and ranking logic for multi-criteria decision analysis.

Experience

ELC Summer Internship, Thapar Institute of Engineering and Technology

🐙 GitHub

June 2025 – July 2025

- Built an ML/DL pipeline to classify ADHD vs. healthy subjects from a 19-channel EEG dataset (121 subjects, ~2.16M records), including preprocessing, stratified splitting, and leakage-free feature scaling.
- Benchmarked Logistic Regression, Random Forest, and a hybrid LSTM+CNN model combining spatial and temporal EEG feature extraction.
- Achieved best results with LSTM+CNN (76.9% accuracy, 0.84 ROC-AUC, 0.85 ADHD recall), outperforming baseline classifiers on non-linear EEG patterns.

Positions of Responsibility

Mess Secretary

Aug 2024 – May 2025

- Served as Mess Secretary of Hostel B, overseeing daily mess operations and coordinating with kitchen and service staff to maintain food quality and service standards.
- Led a 30-member team in planning and executing Hostel Night, overseeing coordination and on-ground execution.